

Lagrangian Formulation — Generalized Coordinates

Classical Mechanics — Lecture Notes

1 Generalized Coordinates

Generalized coordinates are independent variables that uniquely specifies the configuration of a mechanical system.

From the preceding sections we recall that the position of a particle may be written as the Cartesian vector

$$\vec{r} = (x, y, z) \quad (1)$$

or, equally well, in spherical polar coordinates (r, θ, ϕ) or cylindrical coordinates (ρ, ϕ, z) . Any particular mechanical problem can be formulated in terms of many different sets of coordinates.

To specify the position of a single particle in three-dimensional space we need three coordinates: (x, y, z) . For two particles we need six coordinates

$$(x_1, y_1, z_1, x_2, y_2, z_2). \quad (2)$$

For three particles we need nine coordinates, and in general for N particles we need $3N$ coordinates

$$(x_1, y_1, z_1, \dots, x_N, y_N, z_N). \quad (3)$$

As we know, some problems are solved more easily in one coordinate system and some are solved more easily in another. To avoid being specific about the coordinate system we introduce a new set of coordinates called *generalized coordinates*. In this language the position and velocity of a particle are denoted by q_i and $\dot{q}_i$ respectively.

1.1 Transformation Equations

For a specific problem involving one particle we choose generalized coordinate system. We write collectively

$$q_i = (q_1, q_2, q_3). \quad (4)$$

The generalized coordinates are defined as functions of the Cartesian coordinates and of time through a transformation relation:

$$\begin{aligned} q_1 &= q_1(x, y, z, t), \\ q_2 &= q_2(x, y, z, t), \\ q_3 &= q_3(x, y, z, t). \end{aligned} \quad (5)$$

Similarly, for an N -particle system the analogous statements read

$$\begin{aligned} q_1 &= q_1(x_1, y_1, z_1, \dots, x_N, y_N, z_N, t), \\ q_2 &= q_2(x_1, y_1, z_1, \dots, x_N, y_N, z_N, t), \\ &\vdots \\ q_{3N} &= q_{3N}(x_1, y_1, z_1, \dots, x_N, y_N, z_N, t). \end{aligned} \quad (6)$$

Each generalized coordinate depends on all of the Cartesian coordinates and on t through equation (6).

The inverse transformation relations for an N -particle system are

$$\begin{aligned} x_1 &= x_1(q_1, q_2, \dots, q_{3N}, t), \\ y_1 &= y_1(q_1, q_2, \dots, q_{3N}, t), \\ &\vdots \\ z_N &= z_N(q_1, q_2, \dots, q_{3N}, t). \end{aligned} \quad (7)$$

Each Cartesian coordinate depends on all of the generalized coordinates and on time through equation (7).

It is often convenient to denote the whole set of Cartesian coordinates by the single letter x . For a single particle we simply relabel

$$(x, y, z) \longrightarrow (x_1, x_2, x_3). \quad (8)$$

The same convention extends immediately to N particles:

$$\begin{cases} x_1 &= x_1(q_1, q_2, \dots, q_n, t), \\ x_2 &= x_2(q_1, q_2, \dots, q_n, t), \\ &\vdots \\ x_n &= x_n(q_1, q_2, \dots, q_n, t), \end{cases} \quad (9)$$

where $n = 3N$, and

$$\begin{cases} q_1 &= q_1(x_1, x_2, x_3, \dots, x_n, t), \\ q_2 &= q_2(x_1, x_2, x_3, \dots, x_n, t), \\ &\vdots \\ q_n &= q_n(x_1, x_2, x_3, \dots, x_n, t). \end{cases} \quad (10)$$

1.2 Remark

The generalized coordinates q_i , unlike the Cartesian coordinates, do not in general divide into convenient groups of three that can be associated together to form ordinary vectors. Consequently we write the transformation relations equation (9) only in component form:

$$\begin{cases} x_i = x_i(\{q_j\}, t), \\ i = 1, 2, \dots, n, \\ j = 1, 2, \dots, n, \end{cases} \quad (11)$$

and

$$\begin{cases} \vec{r}_i = \vec{r}_i(\{q_j\}, t), \\ i = 1, 2, \dots, N, \\ j = 1, 2, \dots, n. \end{cases} \quad (12)$$

The position vector of the i -th particle retains its usual Cartesian expression

$$\vec{r}_i = x_i \hat{i} + y_i \hat{j} + z_i \hat{k}. \quad (13)$$

1.3 Jacobian Determinant

We adopt the notation

$$(x_1, y_1, z_1, \dots, x_N, y_N, z_N) \equiv (x_1, x_2, \dots, x_n), \quad \text{where } n = 3N. \quad (14)$$

Given the transformation equations from one set of coordinates to another, we can carry out the inverse transformation provided the Jacobian determinant is not zero. Let the q_i be regarded as functions of the x_j . Then

$$J = \frac{\partial(q_1, q_2, \dots, q_n)}{\partial(x_1, x_2, \dots, x_n)} = \begin{vmatrix} \frac{\partial q_1}{\partial x_1} & \frac{\partial q_1}{\partial x_2} & \dots & \frac{\partial q_1}{\partial x_n} \\ \frac{\partial q_2}{\partial x_1} & \frac{\partial q_2}{\partial x_2} & \dots & \frac{\partial q_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial q_n}{\partial x_1} & \frac{\partial q_n}{\partial x_2} & \dots & \frac{\partial q_n}{\partial x_n} \end{vmatrix}. \quad (15)$$

The entries are the partial derivatives of one set of variables with respect to the other. If $\det J \neq 0$ the transformation is locally invertible and the two coordinate systems are equivalent.

2 Generalized Velocity

Consider a mechanical system consisting of a single particle moving in three-dimensional space. We can specify the instantaneous position of this particle using standard Cartesian coordinates, denoted by the set:

$$x_i \in (x_1, x_2, x_3, t) \quad (16)$$

Alternatively, we can describe the configuration of the system using a set of independent parameters known as generalized coordinates, defined as:

$$q_i \in (q_1, q_2, q_3, t) \quad (17)$$

These two sets of coordinates relate to each other through a set of continuous, differentiable transformation equations:

$$x_i = x_i(q_1, q_2, q_3, t) \quad \text{i.e., } x_i = x_i(q_j, t) \quad \text{for } i = 1, 2, 3; j = 1, 2, 3 \quad (18)$$

By explicitly expanding equation (18) for each Cartesian component, we obtain the system of equations:

$$\begin{aligned} x_1 &= x_1(q_1, q_2, q_3, t) \\ x_2 &= x_2(q_1, q_2, q_3, t) \\ x_3 &= x_3(q_1, q_2, q_3, t) \end{aligned} \quad (19)$$

Coordinate System Visualization

To better understand this mathematical mapping, Figure 1 illustrates the coordinate transformation between the physical Cartesian configuration space and the abstract generalized coordinate space.

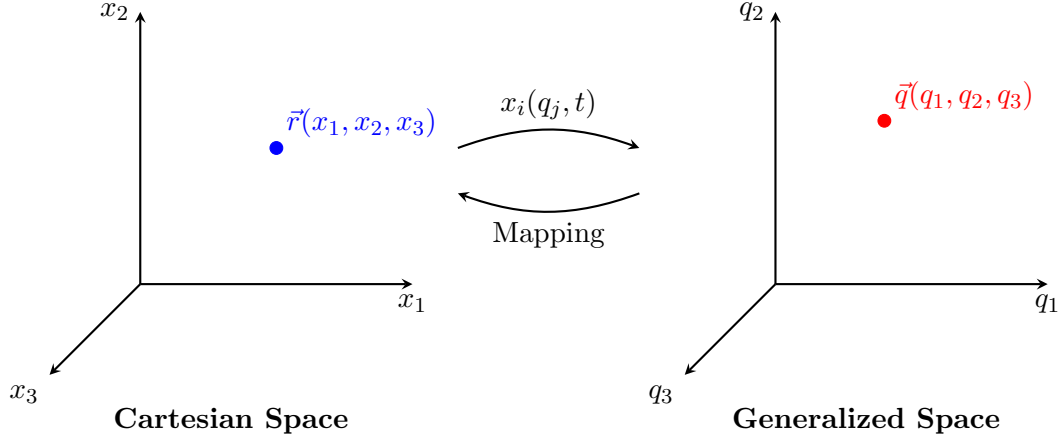


Figure 1: Schematic mapping between standard Cartesian positions and abstract generalized coordinates.

2.1 Derivation of Components of Velocity

We now determine the explicit components of Cartesian velocity in terms of the generalized coordinates and generalized velocities. By definition, the Cartesian velocity components are the total time derivatives of the Cartesian positions:

$$v_i \equiv \frac{dx_i}{dt} \quad \text{where } x_i = x_i(q_1, q_2, q_3, t) \quad (20)$$

Applying the multivariate chain rule allows us to expand the total time derivative with respect to all independent variables:

$$v_i = \frac{\partial x_i}{\partial q_1} \left(\frac{dq_1}{dt} \right) + \frac{\partial x_i}{\partial q_2} \left(\frac{dq_2}{dt} \right) + \frac{\partial x_i}{\partial q_3} \left(\frac{dq_3}{dt} \right) + \frac{\partial x_i}{\partial t} \quad (21)$$

We express this more compactly using summation notation over the coordinate indices:

$$v_i = \sum_{k=1}^3 \frac{\partial x_i}{\partial q_k} \left(\frac{dq_k}{dt} \right) + \frac{\partial x_i}{\partial t} \quad (22)$$

By defining the generalized velocity as the time derivative of the generalized coordinates, $\dot{q}_k \equiv \frac{dq_k}{dt}$, we write the definitive relation for the Cartesian velocity components:

$$v_i = \sum_{k=1}^3 \frac{\partial x_i}{\partial q_k} \dot{q}_k + \frac{\partial x_i}{\partial t} \quad (23)$$

2.2 Relation between Generalized and Cartesian Velocity

To find how the Cartesian velocity components change with respect to the generalized velocities, we take the partial derivative of equation (23) with respect to a specific generalized velocity component $\dot{q}_j$:

$$\frac{\partial v_i}{\partial \dot{q}_j} = \frac{\partial}{\partial \dot{q}_j} \left[\sum_{k=1}^3 \frac{\partial x_i}{\partial q_k} \dot{q}_k \right] + \frac{\partial}{\partial \dot{q}_j} \left(\frac{\partial x_i}{\partial t} \right) \quad (24)$$

Because the coordinate transformation functions $x_i(q, t)$ rely explicitly only on positions and time, their partial derivatives $\frac{\partial x_i}{\partial q_k}$ and $\frac{\partial x_i}{\partial t}$ remain completely independent of the generalized velocities $\dot{q}$. Thus, the second term vanishes entirely:

$$\frac{\partial}{\partial \dot{q}_j} \left(\frac{\partial x_i}{\partial t} \right) = 0 \quad (25)$$

Using the product rule on the remaining summation yields:

$$\frac{\partial v_i}{\partial \dot{q}_j} = \sum_{k=1}^3 \left[\frac{\partial}{\partial \dot{q}_j} \left(\frac{\partial x_i}{\partial q_k} \right) \dot{q}_k + \frac{\partial x_i}{\partial q_k} \left(\frac{\partial \dot{q}_k}{\partial \dot{q}_j} \right) \right] \quad (26)$$

Since $\frac{\partial x_i}{\partial q_k}$ is functionally independent of $\dot{q}_j$, its derivative with respect to $\dot{q}_j$ is identically zero. This is verified by changing the order of differentiation via Clairaut's theorem for mixed partial derivatives:

$$\frac{\partial}{\partial \dot{q}_j} \left(\frac{\partial x_i}{\partial q_k} \right) = \frac{\partial}{\partial q_k} \left(\frac{\partial x_i}{\partial \dot{q}_j} \right) = \frac{\partial}{\partial q_k} (0) = 0 \quad (27)$$

Consequently, the expression simplifies directly to:

$$\frac{\partial v_i}{\partial \dot{q}_j} = \sum_{k=1}^3 \frac{\partial x_i}{\partial q_k} \left(\frac{\partial \dot{q}_k}{\partial \dot{q}_j} \right) \quad (28)$$

Since the generalized velocities are independent variables relative to each other, the derivative $\frac{\partial \dot{q}_k}{\partial \dot{q}_j}$ equals the Kronecker delta symbol, δ_{kj} :

$$\frac{\partial \dot{q}_k}{\partial \dot{q}_j} = \delta_{kj} = \begin{cases} 1 & \text{if } k = j \\ 0 & \text{if } k \neq j \end{cases} \quad (29)$$

Evaluating the summation collapses all terms except for where $k = j$:

$$\frac{\partial v_i}{\partial \dot{q}_j} = \sum_{k=1}^3 \frac{\partial x_i}{\partial q_k} \delta_{kj} = \frac{\partial x_i}{\partial q_j} \quad (30)$$

Equation (30) proves that a Cartesian velocity component relates to a generalized velocity component in the exact same mathematical manner as a Cartesian coordinate relates to a generalized coordinate.

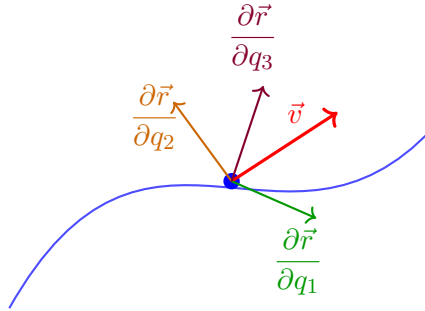


Figure 2: Local decomposition of the velocity vector into contributions proportional to each generalized velocity. The three vectors $\partial \vec{r} / \partial q_k$ form a natural basis at the instantaneous configuration.

Supplementary Information

2.3 Polar to Cartesian Coordinates and the Jacobian

We begin by recalling the standard transformation between polar coordinates (r, θ) and Cartesian coordinates (x, y) . The forward map from polar to Cartesian reads

$$x = r \cos \theta, \quad (31)$$

$$y = r \sin \theta. \quad (32)$$

The inverse map is

$$r = (x^2 + y^2)^{1/2}, \quad (33)$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right), \quad (34)$$

where the arctangent must be evaluated with care to recover the correct quadrant.

The Jacobian matrix of the transformation from the Cartesian pair (x, y) to the polar pair (r, θ) is defined by

$$J = \frac{\partial(r, \theta)}{\partial(x, y)} = \begin{pmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{pmatrix}. \quad (35)$$

We compute each partial derivative explicitly. First,

$$\frac{\partial r}{\partial x} = \frac{\partial}{\partial x}(x^2 + y^2)^{1/2} = \frac{1}{2}(x^2 + y^2)^{-1/2} \cdot 2x = \frac{x}{r}. \quad (36)$$

Likewise,

$$\frac{\partial r}{\partial y} = \frac{y}{r}. \quad (37)$$

For the angular derivatives we start from $\theta = \tan^{-1}(y/x)$. Differentiating with respect to x yields

$$\frac{\partial \theta}{\partial x} = \frac{1}{1 + (y/x)^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{1}{(1 + (y^2/x^2))} \cdot \left(-\frac{y}{x^2}\right) = \frac{x^2}{x^2 + y^2} \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2 + y^2} = -\frac{y}{r^2}. \quad (38)$$

Differentiating with respect to y gives

$$\frac{\partial \theta}{\partial y} = \frac{1}{1 + (y/x)^2} \cdot \frac{1}{x} = \frac{x^2}{x^2 + y^2} \cdot \frac{1}{x} = \frac{x}{r^2}. \quad (39)$$

Assembling the matrix, we obtain

$$J = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{pmatrix}. \quad (40)$$

The determinant is

$$\det J = \left(\frac{x}{r}\right)\left(\frac{x}{r^2}\right) - \left(\frac{y}{r}\right)\left(-\frac{y}{r^2}\right) = \frac{x^2}{r^3} + \frac{y^2}{r^3} = \frac{x^2 + y^2}{r^3} = \frac{r^2}{r^3} = \frac{1}{r}. \quad (41)$$

Thus

$$J = \frac{1}{r} \neq 0 \quad (42)$$

provided $r \neq 0$. The non-vanishing of the Jacobian guarantees that the transformation is locally invertible away from the origin.

Under a change of variables the area element transforms according to

$$dx dy = |\det J_{(x,y) \leftarrow (r,\theta)}| dr d\theta. \quad (43)$$

The inverse Jacobian (from polar to Cartesian) has determinant r , so the familiar polar area element is recovered:

$$dx dy = r dr d\theta. \quad (44)$$

Equivalently, the notes write the relation in the opposite direction,

$$dr d\theta = \frac{1}{r} dx dy, \quad (45)$$

which is consistent with $|J| = 1/r$.

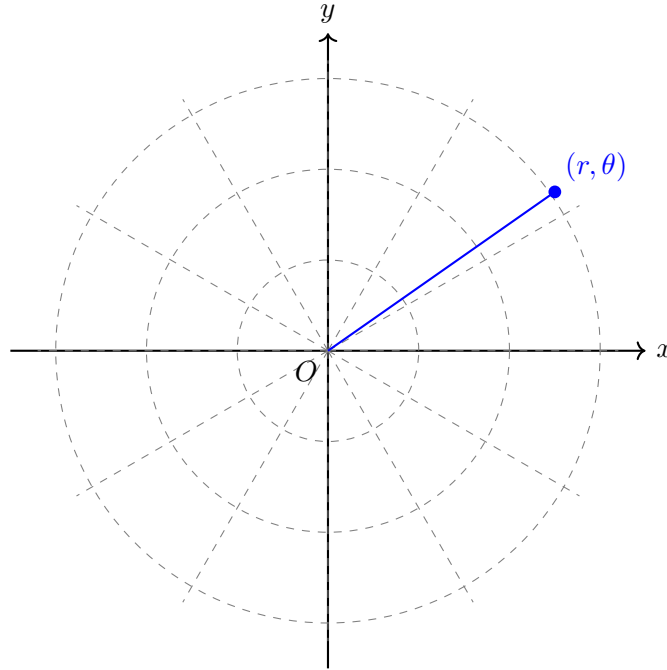


Figure 3: Polar coordinate grid superimposed on the Cartesian plane. The Jacobian determinant $1/r$ quantifies the local area distortion between the two coordinate systems.

2.4 Extension to Three Dimensions and Spherical Coordinates

We now consider the analogous transformation in three-dimensional space. Let

$$r = (x^2 + y^2 + z^2)^{1/2}, \quad (46)$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right), \quad (47)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right). \quad (48)$$

(The precise definition of the polar angle θ may vary by convention; here we follow the notes.)

The Jacobian matrix of the map $(x, y, z) \mapsto (r, \theta, \phi)$ is the 3×3 matrix whose entries are all first partial derivatives

$$J = \begin{pmatrix} \partial r / \partial x & \partial r / \partial y & \partial r / \partial z \\ \partial \theta / \partial x & \partial \theta / \partial y & \partial \theta / \partial z \\ \partial \phi / \partial x & \partial \phi / \partial y & \partial \phi / \partial z \end{pmatrix}. \quad (49)$$

We evaluate the radial derivatives first:

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}. \quad (50)$$

For the polar-angle derivatives the notes give

$$\frac{\partial \theta}{\partial x} = \frac{\partial}{\partial x} \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right) = \frac{1}{1 + (\sqrt{x^2 + y^2}/z)^2} \cdot \frac{\partial}{\partial x} \left(\frac{\sqrt{x^2 + y^2}}{z} \right). \quad (51)$$

After simplification one obtains expressions proportional to

$$\frac{\partial \theta}{\partial x} \sim \frac{xz}{r^2 \sqrt{x^2 + y^2}}, \quad \frac{\partial \theta}{\partial y} \sim \frac{yz}{r^2 \sqrt{x^2 + y^2}}, \quad \frac{\partial \theta}{\partial z} = -\frac{\sqrt{x^2 + y^2}}{r^2}. \quad (52)$$

(The exact algebraic factors appear in the handwritten notes and are recovered by straightforward, albeit lengthy, differentiation.)

The azimuthal derivatives are identical to the two-dimensional case:

$$\frac{\partial \phi}{\partial x} = -\frac{y}{x^2 + y^2}, \quad \frac{\partial \phi}{\partial y} = \frac{x}{x^2 + y^2}, \quad \frac{\partial \phi}{\partial z} = 0. \quad (53)$$

When the full determinant is computed, the Jacobian factor that converts the Cartesian volume element into spherical coordinates is

$$dx \, dy \, dz = r^2 \sin \theta \, dr \, d\theta \, d\phi. \quad (54)$$

(The notes contain intermediate algebraic identities that verify this classical result step by step; each identity follows from the product rule and the relation $r^2 = x^2 + y^2 + z^2$.)

Remark

All partial derivatives are obtained by the chain rule. The vanishing of $\partial \phi / \partial z$ is expected on physical grounds: the azimuthal angle is independent of height. The factor $r^2 \sin \theta$ arises because spherical shells grow with r^2 while the polar circles shrink as $\sin \theta$.

Lagrangian Formulation — Generalized Coordinates

Classical Mechanics — Lecture Notes

1 Degrees of Freedom and Constraints

Every physical system possesses a characteristic number of **degrees of freedom** (DoF). We define this quantity as the minimum number of independent coordinates required to uniquely specify the exact configuration of every particle in the system at any given moment.

1.1 Free Particles

- **Single Free Particle:** A lone particle moving unrestricted in three-dimensional physical space requires three independent coordinates to pinpoint its position. We can specify this location using Cartesian coordinates (x, y, z) , spherical coordinates (r, θ, ϕ) , or cylindrical coordinates (ρ, ϕ, z) . Therefore, its total degrees of freedom equal exactly 3:

$$\text{DoF} = 3 \tag{1}$$

- **System of N Free Particles:** If a mechanical system comprises N entirely unconstrained, independent particles, each individual particle possesses 3 independent spatial degrees of freedom. Consequently, the entire system requires $3N$ independent coordinates to locate every point simultaneously:

$$\text{DoF} = 3N \tag{2}$$

1.2 Constrained Motion

When a physical system faces geometric restrictions or mechanical bounds, we say the system is acted upon by forces of constraint. These physical restrictions introduce **mathematical dependencies among the coordinates**, allowing us to **reduce** the total number of parameters needed to describe the system's overall motion. As a result, constraints systematically decrease the degrees of freedom available to the system.

1.2.1 Example: A Particle Rotating in a Plane

Consider a single particle restricted to a flat two-dimensional plane (xy -plane), anchored to a central origin by a rigid, inextensible rod of fixed length r .

$$z = \text{constant} \quad (\text{Restricted to a 2D plane}) \tag{3}$$

$$r = \text{fixed length } l \quad (\text{Rigid constraint}) \tag{4}$$

Mathematically, if we begin with a 3D description (x, y, z) , we apply two independent constraint equations ($z = c$ and $x^2 + y^2 = l^2$). Thus, the system reduces from 3 original coordinates down to a single remaining independent coordinate, namely the angular parameter ϕ . The system possesses exactly one degree of freedom.

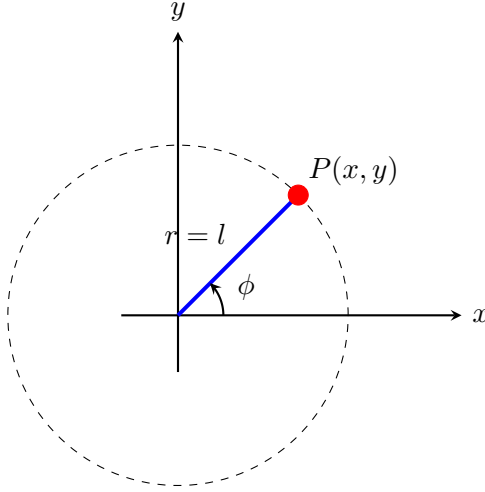


Figure 1: A particle confined to rotate along a circular pathway of fixed radius r in a plane.

1.3 Additional Geometry-Driven Constraint Examples

1.3.1 Unconstrained Planar Motion

If a particle moves freely within a two-dimensional flat plane without radial boundaries, we freeze its vertical component ($z = \text{constant}$).

$$\text{Number of Constraints } (k) = 1 \quad (5)$$

The original 3D system loses one degree of freedom, meaning that a set of two coordinates—either Cartesian (x, y) or polar (r, ϕ) —completely describes the particle's state:

$$\text{DoF} = 3 - 1 = 2 \quad (6)$$

1.3.2 The Plane Pendulum

Consider a simple (planar) pendulum oscillating under a rigid ceiling support line where $z = 0$. The particle hangs from a fixed pivot point via an unyielding string of length l , satisfying $x^2 + y^2 = l^2 = \text{constant}$.

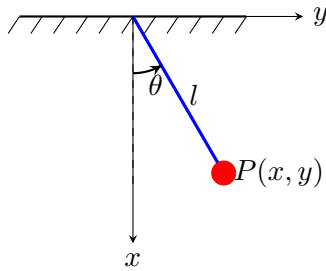


Figure 2: Coordinate structure of a single-link simple planar pendulum.

These combined limits constitute two explicit coordinate restrictions:

$$z = 0 \quad \text{and} \quad x^2 + y^2 = l^2 \quad (7)$$

We expand the local variables using trigonometry:

$$x = l \cos \theta, \quad y = l \sin \theta \quad (8)$$

Consequently, knowing the isolated angular displacement value $\theta = \tan^{-1}(y/x)$ is entirely sufficient to locate the oscillating particle P at any instant.

1.3.3 Particle Restrained to a Spherical Shell Surface

Let a particle move along the outer surface of a fixed solid sphere of static radius r .

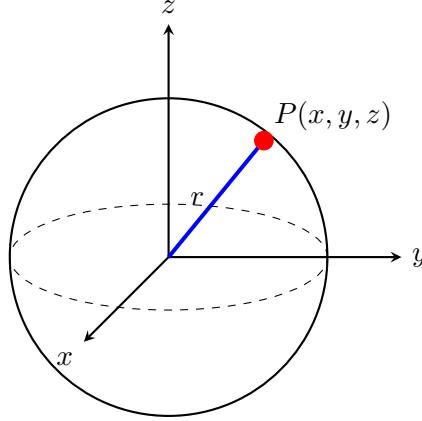


Figure 3: A particle tracking exclusively across a spherical surface shell.

We define this single geometric constraint by the algebraic relation:

$$x^2 + y^2 + z^2 = r^2 \implies x^2 + y^2 + z^2 - r^2 = 0 \quad (9)$$

This system possesses exactly one constraint ($k = 1$), leaving two active degrees of freedom. We can uniquely identify every point on the shell using the two independent angular parameters of a spherical coordinate framework, (θ, ϕ) .

1.4 General Fundamental Rule for Total Degrees of Freedom

If a broad physical assembly containing N individual particles encounters k separate, independent geometric constraint equations, we calculate the remaining net degrees of freedom and the number of independent coordinates as:

$$\text{DoF} = n = 3N - k \quad (10)$$

2 Classification of Constraints

2.1 Holonomic and Non-Holonomic Constraints

We classify physical constraints based on how their equations depend on coordinate and velocity parameters.

1. **Holonomic Constraints:** We define a constraint as holonomic if we can express its limiting relationship purely as an algebraic equality relating positional coordinates and time:

$$f(q_1, q_2, q_3, \dots, q_n, t) = 0 \quad (11)$$

where $n = 3N - k$ matches the net independent generalized coordinates q_i assigned to the configuration structure. Crucially, holonomic constraint relations are completely independent of velocity parameters, or can be integrated to become independent of velocity.

2. **Non-Holonomic Constraints:** If we cannot express a coordinate limitation as a direct algebraic equality, or if the relation depends fundamentally on non-integrable velocity components, the constraint is non-holonomic. These equations are often written as inequalities (e.g., a particle trapped inside a sphere of radius r , written as $x^2 + y^2 + z^2 - r^2 \geq 0$) and cannot be simplified to exclude velocity dependencies.

2.1.1 Example of an Integrable Holonomic System: Rolling Disc

Consider a uniform solid disc of radius a rolling down an inclined plane along its line of steepest slope. We track the linear displacement along the incline, denoted by l , and the rotational angle, denoted by θ .

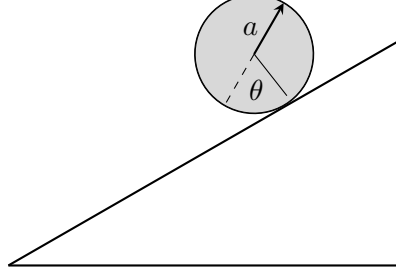


Figure 4: A rolling disc system exhibiting an integrable velocity relationship.

We begin by stating the continuous, non-slipping constraint matching linear velocity to angular velocity:

$$\frac{dl}{dt} = a \frac{d\theta}{dt} \implies \dot{l} = a\dot{\theta} \quad (12)$$

We can integrate this differential velocity relationship directly with respect to time:

$$\int dl = \int a d\theta \implies l = a\theta + \text{constant} \quad (13)$$

Rearranging terms yields a standard algebraic expression:

$$l - a\theta = \text{constant} \quad (14)$$

Because we successfully integrated the velocity components into a pure coordinate relationship, this rolling constraint is fundamentally **holonomic**.

2.1.2 Example: Rigid Body Configuration Space

A rigid body represents a classic example of a holonomic system. For any two internal particles i and j within the body, the spatial distance between them remains constant throughout all motion:

$$|\vec{r}_i - \vec{r}_j| = \vec{r}_i \cdot \vec{r}_j = \text{constant} = C_{ij} \quad (15)$$

Squaring both sides eliminates the radical vector norm, yielding an algebraic equality:

$$\vec{r}_i \cdot \vec{r}_j - C_{ij}^2 = 0 \quad (16)$$

Let us calculate the net degrees of freedom for an arbitrary rigid body composed of N constituent particles:

1. We can place the first particle (x_1, y_1, z_1) anywhere in unconstrained space, which consumes exactly 3 spatial coordinates (DoF1 = 3).
2. We place the second particle (x_2, y_2, z_2) at a fixed distance r_{12} from the first particle, introducing 1 constraint equation ($r_{12} = \text{fixed}$). This leaves 2 independent coordinates (DoF2 = 2).
3. We position the third particle (x_3, y_3, z_3) relative to the first two, which introduces 2 independent distance constraints ($r_{23} = \text{fixed}$ and $r_{13} = \text{fixed}$). This leaves only 1 degree of freedom (DoF3 = 1).

4. For every subsequent particle $k \geq 4$, its position is completely locked in place relative to the first three points via 3 separate rigid distance constraints. Thus, no additional degrees of freedom are introduced ($\text{DoF}_k \geq 4 = 0$).

Summing these independent components yields the total degrees of freedom for any standard three-dimensional rigid body:

$$\text{DoF} = 3 + 2 + 1 + 0 + \cdots + 0 = 6, \quad (17)$$

which includes three translational and three rotational degrees of freedom.

2.2 Scleronomous and Rheonomous Constraints

We can further classify constraints based on whether they depend explicitly on time.

- **Scleronomous Constraints:** These constraints do not depend explicitly on time. Their governing mathematical relations depend purely on spatial coordinates:

$$f(q_1, q_2, \dots, q_n) = 0 \quad (18)$$

Classic examples include rigid bodies, simple pendulums with fixed supports, and standard static spherical shells.

- **Rheonomous Constraints:** These constraints depend explicitly on time. Their mathematical governing relations take the general form:

$$f(q_1, q_2, \dots, q_n, t) = 0 \quad (19)$$

Examples include a simple pendulum with a moving suspension point, a bead sliding along a actively rotating wire, or a balloon whose radius changes over time ($r = r(t)$).

3 Implicit and Explicit Time Dependence

We must distinguish between variables that change over time implicitly (mediated via positions or velocities) and those that depend on time explicitly.

3.1 Case 1: Pure Implicit Dependence

Consider a static potential energy function U that depends solely on geometric position coordinates:

$$U = U(x, y, z) \quad (20)$$

Taking its total derivative using the multivariate chain rule yields:

$$dU = \frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial y} dy + \frac{\partial U}{\partial z} dz \quad (21)$$

Because time t does not appear explicitly inside the functional definition of U , its explicit partial time derivative is identically zero:

$$\frac{\partial U}{\partial t} = 0 \quad (22)$$

3.2 Case 2: Explicit Time Dependence

Now consider a time-varying potential field U that changes dynamically over time, even at a fixed location:

$$U = U(x, y, z, t) \quad (23)$$

Its total differential expansion must include an explicit time derivative term:

$$dU = \frac{\partial U}{\partial x}dx + \frac{\partial U}{\partial y}dy + \frac{\partial U}{\partial z}dz + \frac{\partial U}{\partial t}dt \quad (24)$$

In this case, the explicit partial derivative with respect to time is non-zero:

$$\frac{\partial U}{\partial t} \neq 0 \quad (25)$$

4 Worked Example: Mathematical and Geometric Characterization

4.1 Pendulum Constraints under Stationary and Driving Conditions

4.1.1 Scenario A: Fixed Support Origin

Consider a simple pendulum of length l hanging from a fixed, stationary pivot point O . The

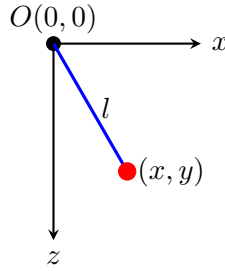


Figure 5: A simple pendulum suspended from a fixed, stationary origin.

path of the pendulum bob is constrained by the length of the string:

$$\sqrt{x^2 + y^2} = l \implies x^2 + y^2 = l^2 \implies x^2 + y^2 - l^2 = 0 \quad (26)$$

Because this relation relies purely on spatial positions and does not contain time explicitly, it constitutes a classic **holonomic, scleronomous constraint**.

4.1.2 Scenario B: Oscillating Support Origin

Now suppose the suspension pivot point O is not stationary, but instead drives horizontally in harmonic motion over time:

$$x'(t) = x_0 \cos \omega t \quad (27)$$

This horizontal movement shifts the relative position coordinate of the pendulum bob:

$$x_{\text{new}} = x - x'(t) = x - x_0(1 - \cos \omega t) \quad (28)$$

Substituting this shifting coordinate framework back into our geometric length constraint equation yields:

$$[x - x_0(1 - \cos \omega t)]^2 + y^2 - l^2 = 0 \quad (29)$$

Time now appears explicitly inside the constraint equation, so the constraint is still **holonomic but rheonomic**.

5 Detailed Exercises and Solutions

1. Particle inside a cylindrical surface.

- (a) The particle remains at a fixed distance R from the vertical axis of the cylinder. In Cartesian coordinates the constraint is therefore

$$x^2 + y^2 = R^2. \quad (30)$$

- (b) A free particle in three-dimensional space has three degrees of freedom. The single independent holonomic constraint removes one of them, leaving

$$\text{DoF} = 3 - 1 = 2. \quad (31)$$

- (c) A convenient choice of generalized coordinates is the vertical height z and the azimuthal angle ϕ . The Cartesian coordinates are recovered by

$$x = R \cos \phi, \quad (32)$$

$$y = R \sin \phi, \quad (33)$$

$$z = z. \quad (34)$$

- (d) The constraint equation does not contain the time t explicitly. It is therefore scleronomic.

2. Rigid dumb-bell.

- (a) Each of the two point masses would contribute three degrees of freedom if free, giving a total of six. The single rigid-length constraint

$$|\vec{r}_1 - \vec{r}_2| = \ell \quad (35)$$

removes one degree of freedom. Equivalently one may count three translational degrees of freedom of the centre of mass plus two rotational degrees of freedom (the direction of the axis of the dumb-bell is specified by two angles). Hence

$$\text{DoF} = 5. \quad (36)$$

- (b) When the dumb-bell is confined to a fixed plane an additional constraint (the vanishing of the z -coordinates of both masses, or equivalently the requirement that the axis lie in the plane) appears. The count falls to three:

$$\text{DoF} = 3. \quad (37)$$

This includes two translational degrees of freedom of the centre of mass inside the plane and one rotational degree of freedom about the axis perpendicular to the plane.

3. Particle on an ellipsoid.

- (a) The given surface equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad (38)$$

is a finite relation among the Cartesian coordinates alone. It is therefore holonomic. The time t does not appear, so the constraint is also scleronomic.

- (b) One independent holonomic constraint reduces the original three degrees of freedom to two:

$$\text{DoF} = 3 - 1 = 2. \quad (39)$$

- (c) Two convenient generalized coordinates are the elliptic angles (ϑ, φ) defined by

$$x = a \sin \vartheta \cos \varphi, \quad (40)$$

$$y = b \sin \vartheta \sin \varphi, \quad (41)$$

$$z = c \cos \vartheta. \quad (42)$$

4. Double pendulum.

- (a) Each pendulum bob would contribute two planar degrees of freedom if free, giving four in total. Two independent length constraints reduce the count by two, leaving

$$\text{DoF} = 2. \quad (43)$$

- (b) A natural pair of generalized coordinates is the pair of angles (θ_1, θ_2) that the two rods make with the downward vertical.

- (c) The holonomic constraints that keep the rods rigid are

$$x_1^2 + y_1^2 = \ell_1^2, \quad (44)$$

$$(x_2 - x_1)^2 + (y_2 - y_1)^2 = \ell_2^2. \quad (45)$$

- (d) In terms of the angles the Cartesian coordinates read

$$x_1 = \ell_1 \sin \theta_1, \quad (46)$$

$$y_1 = -\ell_1 \cos \theta_1, \quad (47)$$

$$x_2 = \ell_1 \sin \theta_1 + \ell_2 \sin \theta_2, \quad (48)$$

$$y_2 = -\ell_1 \cos \theta_1 - \ell_2 \cos \theta_2. \quad (49)$$

5. Spherical pendulum.

- (a) The rigid-rod constraint is simply

$$x^2 + y^2 + z^2 = \ell^2. \quad (50)$$

- (b) One holonomic constraint leaves two degrees of freedom.

- (c) Introduce the spherical polar angles (θ, ϕ) by

$$x = \ell \sin \theta \cos \phi, \quad (51)$$

$$y = \ell \sin \theta \sin \phi, \quad (52)$$

$$z = \ell \cos \theta. \quad (53)$$

6. Time-dependent sphere.

- (a) The particle remains on the surface of a sphere whose radius grows linearly with time:

$$x^2 + y^2 + z^2 = (R_0 + \alpha t)^2. \quad (54)$$

- (b) The equation is a finite relation among the coordinates and the time; it is therefore holonomic. Because t appears explicitly the constraint is rheonomic.

- (c) One holonomic constraint leaves two degrees of freedom.
- (d) Yes. Locally one may solve the constraint for any one of the Cartesian coordinates (for example z) in terms of the other two coordinates and the time, provided the corresponding partial derivative does not vanish. The remaining two coordinates then serve as independent generalized coordinates.